D 142 | Attribution is not causal lift

PRIVATE FICTIONAL offline channel lab. Rupees; one acquisition cohort and separate geo period. No ad accounts, inventory inspection, analytics changes or spend.

One cohort	100000 impressions, CPM 60, CTR.02, purchase rate.05 per click =>spend 6000, clicks 2000, new customers 100. Each purchase is stipulated distinct new customer. AOV 300, first-order contribution fraction.4 =>revenue 30000, contribution 12000, net after acquisition 6000, CAC 60.
Separate geo toy	Treatment orders 1000 pre/1250 post, control 800 pre/880 post. Assumed proportional counterfactual 1000*(880/800)=1100; incremental 150. AOV 300, unit contribution 120, incremental spend 20000 =>iROAS 2.25, net-2000. Assumptions not validated by arithmetic.
Counterfactual sensitivity	Additive parallel-change assumption predicts 1080 and lift 170, not 150. It would give net 400. This is a different identifying assumption, not a correction or a choice to maximize results. Scenario factor 1 scales proportional counterfactual only; not a measured contamination correction.
Revenue is not contribution	At margin.4, iROAS break-even 2.5, not 1. Incremental-order break-even 166.666667, whole 167. Positive revenue lift can lose money. Margin consistent with 120/300 only in baseline; independent unit contribution remains explicitly supplied.
Illustrative bounds	Supplied toy lift endpoints-20 and 320, no confidence level or estimation method. They map linearly to iROAS-.3 and 4.8, net-22400 and 18400. These are scenario endpoints, not a computed confidence interval or decision to spend. Negative lift retained.
Observation and action	30% brand-search last-click is allocation, not 30% causal sales. Ad longevity, creative and density do not reveal CAC, profit, spend or channel mix. Actual inventory/evidence/consent/owner/stop plan blank. No real 10-lakh allocation, shutdown, tracking integration or overall 10.

Inputs | Independent cohort and geo families

All values fictional. Fractions not percentages. Cohort buyers may be fractional expectations when inputs change, not actual people counts. Geo orders whole.

Cohort impressions	100000.00
CPM rupees per thousand impressions	60.00
Click-through fraction	0.02
Purchase fraction per click	0.05
Cohort AOV rupees	300.00
First-order contribution fraction	0.40

Inputs | Independent cohort and geo families

All values fictional. Fractions not percentages. Cohort buyers may be fractional expectations when inputs change, not actual people counts. Geo orders whole.

Treatment pre orders	1000.00
Treatment post orders	1250.00
Control pre orders	800.00
Control post orders	880.00
Geo AOV rupees	300.00
Geo incremental unit contribution rupees	120.00
Geo incremental spend rupees	20000.00
Counterfactual scenario multiplier	1.00
Supplied illustrative lower lift orders	-20.00
Supplied illustrative upper lift orders	320.00

Cohort numeric gate (1=yes)	1.000000
Geo numeric gate (1=yes)	1.000000
Supplied bounds ordered (1=yes)	1.000000

Signed endpoints allowed. Bounds do not estimate uncertainty or require a point estimate to lie inside. Numeric gates do not validate randomization, parallel trends, power, exposure, spillovers, conversion lag, margin completeness or permissions. Zero denominators remain undefined.

Acquisition | First-order economics

Authored conditional arithmetic only. Rupees except labeled fractions, ratios or order counts.

Spend	6000.000000
Clicks	2000.000000
Distinct new customers under assumption	100.000000
Customer acquisition cost	60.000000
First-order revenue	30000.000000
Unit contribution before acquisition	120.000000
Cohort contribution before acquisition	12000.000000
Cohort contribution after acquisition	6000.000000
Attributed first-order revenue / spend	5.000000
Contribution per acquired customer, after acquisition	60.000000

Baseline 6000, 2000, 100, 60, 30000, 120, 12000, 6000, 5, 60. Attribution and distinct-customer labels are assumptions, not provider observations. No repeat profits, LTV, churn, other fixed costs or causal incremental channel effect estimated. Zero purchases leaves CAC undefined, not free acquisition.

Geo | Two different counterfactual assumptions

Authored conditional arithmetic only. Rupees except labeled fractions, ratios or order counts.

Control post / pre trend ratio	1.100000
Proportional expected treatment post	1100.000000
Proportional incremental orders	150.000000
Incremental revenue	45000.000000
Incremental revenue / incremental spend (iROAS)	2.250000
Incremental contribution after spend	-2000.000000
Revenue break-even iROAS at supplied unit margin	2.500000
Incremental order break-even, continuous	166.666667
Incremental order break-even, whole	167.000000
Additive expected treatment post	1080.000000
Additive incremental orders	170.000000
Additive contribution after spend	400.000000

Baseline proportional 1.1, 1100, 150, 45000, 2.25, -2000; revenue break-even 2.5, orders 166.666667/167. Additive 1080, 170, 400 is a different assumption. Neither is automatic causal evidence from two states/four weeks. Negative lift/net retained; no fitted synthetic control, randomization, power or variance estimate.

Sensitivity | Scenario, not estimated interval

Authored conditional arithmetic only. Rupees except labeled fractions, ratios or order counts.

Scenario expected treatment post	1100.000000
Scenario incremental orders	150.000000
Scenario incremental revenue	45000.000000
Scenario iROAS	2.250000
Scenario contribution after spend	-2000.000000

Multiplier 1 restores proportional result. Factor.9 gives expected 990, lift 260, net 11200; 1.1 gives 1210, lift 40, net-15200. Choosing favorable assumptions after results is not valid design. This multiplier is not contamination correction or a probability distribution.

Bounds | Linear mapping of supplied endpoints

Authored conditional arithmetic only. Rupees except labeled fractions, ratios or order counts.

Supplied lower lift orders	-20.000000
Supplied upper lift orders	320.000000
Lower incremental revenue	-6000.000000
Upper incremental revenue	96000.000000
Lower iROAS	-0.300000
Upper iROAS	4.800000
Lower contribution after spend	-22400.000000
Upper contribution after spend	18400.000000

Baseline-20, 320, -6000, 96000, -.3, 4.8, -22400, 18400. Endpoints entirely stipulated, no confidence level, standard error or estimator. Fixed positive AOV/unit contribution and spend assumed for mapping. Real interval crossing zero is not proof of zero effect; decision precision and economic hurdle differ.

Design plan | Recommendation without activation

No real experiment, holdout shutdown, ad inventory research, API integration, measurement change or 10-lakh allocation.

Observable versus inferred	If later authorized, record dated ad IDs, creative, format, offer, source and coverage. Longevity and ad density are observations, not spend, CAC, profit or actual channel mix. Meta overview returned only a short EU/branded-content paragraph; do not claim full global disclosure rules checked.
Attribution versus lift	30%last-click brand-search revenue describes allocation only. Define actual counterfactual question and experimental unit; do not relabel attributed conversions as unbiased causal outcomes. Treatment means assigned intervention, not always ad exposure: go-dark treatment is unexposed to the tested ads.
Feasibility and timing	Google says access is not universal. Resolve compatible outcomes, country/units, preperiod fit, power/minimum detectable effect, purchase cycle, learning and conversion lag. Two states/four weeks is not enough by definition. No account-team contact or settings changes from this lab.
Contamination and integrity	Check cross-region travel, targeting, other media, overlapping studies, geography changes and KPI consistency. Preserve prespecified assignment and stop rules, not switching losing geos after seeing results. Current Google guidance is a source, not approval to run its operational steps.
Economics and uncertainty	Require contribution-adjusted hurdle, not merely iROAS>1. Predeclare interval/precision and downside rules; wide interval spanning zero is inconclusive at that precision, not certain zero. Supplied bounds here are scenarios, not a fabricated confidence interval.
Transfer check	Close notes: halve CTR and explain spend unchanged/buyers 50/CAC 120/net 0; double geo unit contribution and explain iROAS unchanged/net 16000; change control post to 1000 and explain proportional lift 0/net-20000. Actual learner responses blank; historical Airbnb-Craigslist not current permission or scraping recipe.

Actual plan | Inventory, evidence and approval blank

No real brand ad inventory, geo data, account access, budget, experiment, participant consent or accepted owner.

Question / KPI	Source / date	Unit / preperiod	Assumption gap	Owner / acceptance	Approval / budget	Stop / review

Only record real evidence if later authorized. Blank unknowns do not become zero lift or approval. No contact, ad spend, holdout activation, analytics integration, API or account change. No independent learner, full source/AV/native Excel/Sheets parity or overall 10.

Evidence | Sources do not execute experiments

Six URLs from four organizations; official measurement guidance, practitioner history and accounting terms. Not six independent effectiveness studies.

Google setup	https://support.google.com/google-ads/answer/14097193?hl=en | Complete returned written page read: eligibility, feasibility, units/contamination and setup. Not an account-specific eligibility check or permission to change settings. Source advice to increase budgets/contact representatives not acted on.
Google metrics	https://support.google.com/google-ads/answer/14102986?hl=en | Complete returned written page read: iROAS incremental value/cost, counts/cost, cooldown, statuses/interval. Revenue-based efficiency wording not contribution-profit guarantee. Post hoc positive-slice suggestion not adopted as confirmatory evidence.
Google Meridian	https://developers.google.com/meridian/geox/implementation-of-geo-testing | Complete returned written guidance/table read: units, KPI, purchase cycle, cooldown, heavy-up/go-dark/holdback, isolation and targeting. Its 7-14 week example not universal schedule. Figure pixels and linked methodologies not read; no implementation.
Meta overview	https://transparency.meta.com/en-us/researchtools/ad-library-tools | Only returned heading/updated 2023 and short EU/branded-content paragraph read. Not full global policy/API record fields or actual live ad search. No competitor inventory/performance asserted.
Andrew Chen history	https://andrewchen.com/how-to-be-a-growth-hacker-an-airbnbcraigslist-case-study/ | Complete returned written historical case read, not images. Dated reach and reverse-engineering narrative not verified current facts, permission or workflow to reproduce. No integrations/scraping/pixels performed.
OpenStax accounting	https://openstax.org/books/principles-managerial-accounting/pages/3-key-terms | Prior D 141 complete returned written keyterms reused: contribution, break-even, relevant range. No expanded chapter/book coverage. No source diagram/podcast AV, real experiment/independent learner/native cross-viewer/account/overall 10 clearance.

Failure log | Authored checks, not campaign results

Execution log filled after testing. No causal validation, real customer evidence, account test or independent learner.

1	PASS: 38 baseline formulas independently recomputed.
2	PASS: 24 changed sets plus restore.
3	PASS: 16 invalid economics/608 isolation checks.
4	PASS: 4 invalid bounds/152 isolation checks.
5	PASS: 6 broken references/6 explicit checks.
6	PASS: independent cohort/geo/bounds isolation.
7	PASS: zero denominators undefined;negative lift retained.
8	PASS: actual evidence/approval/budget plan blank.